Diseases of turmeric

1. Taphrina leaf blotch - *Taphrina maculans*

2. Leaf spot or anthracnose - *Colletotrichum capsici*

3. Rhizome rot or root rot - *Pythium aphanidermatum*, *P. graminicolum* & *Fusarium solani*

Taphrina leaf blotch - *Taphrina maculans*

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Leaves become distorted and appear **reddish brown**

Taphrina leaf blotch - *Taphrina maculans*

- This disease was first described by **Butler (1911)** and now it has been observed in all the turmeric growing areas of the country.

- Severe outbreaks of this disease have been **reported from Rayalaseema area of Andhra Pradesh.**

Symptoms:

- Spots of **1-2 mm in diameter arranged in rows along the veins**, first appear as **pale yellow discolouration** not sharply defined from the rest of the tissue, become **dirty yellow** and then deepen to the **colour of old gold and some times to bay shade.**

- **First on lower leaves and spread to upper leaves in October and November.**

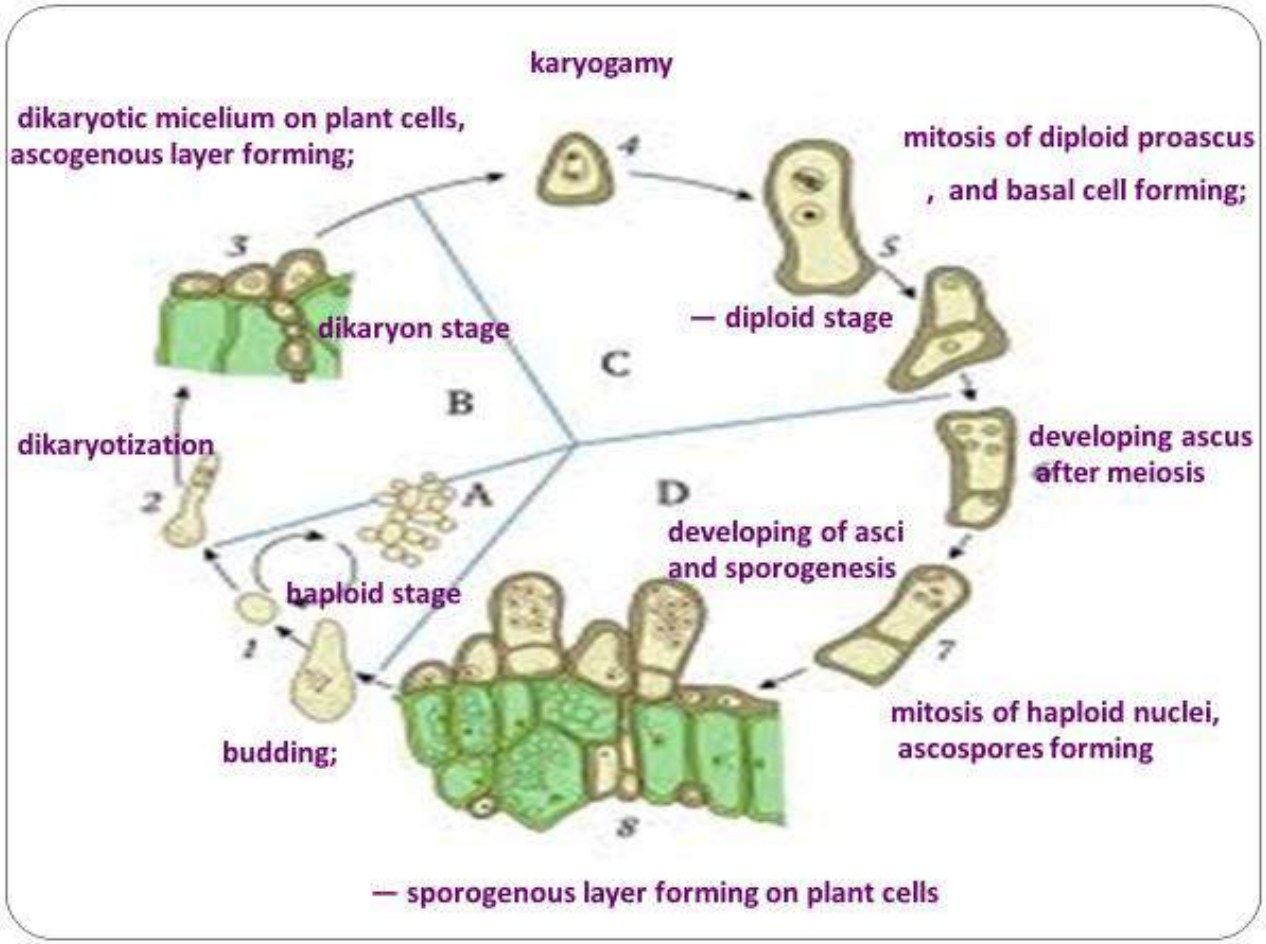
- Several spots of the disease appear on **both surfaces of leaves, being generally more numerous on the upper surface**.

Taphrina leaf blotch - *Taphrina maculans*

Symptoms contd…

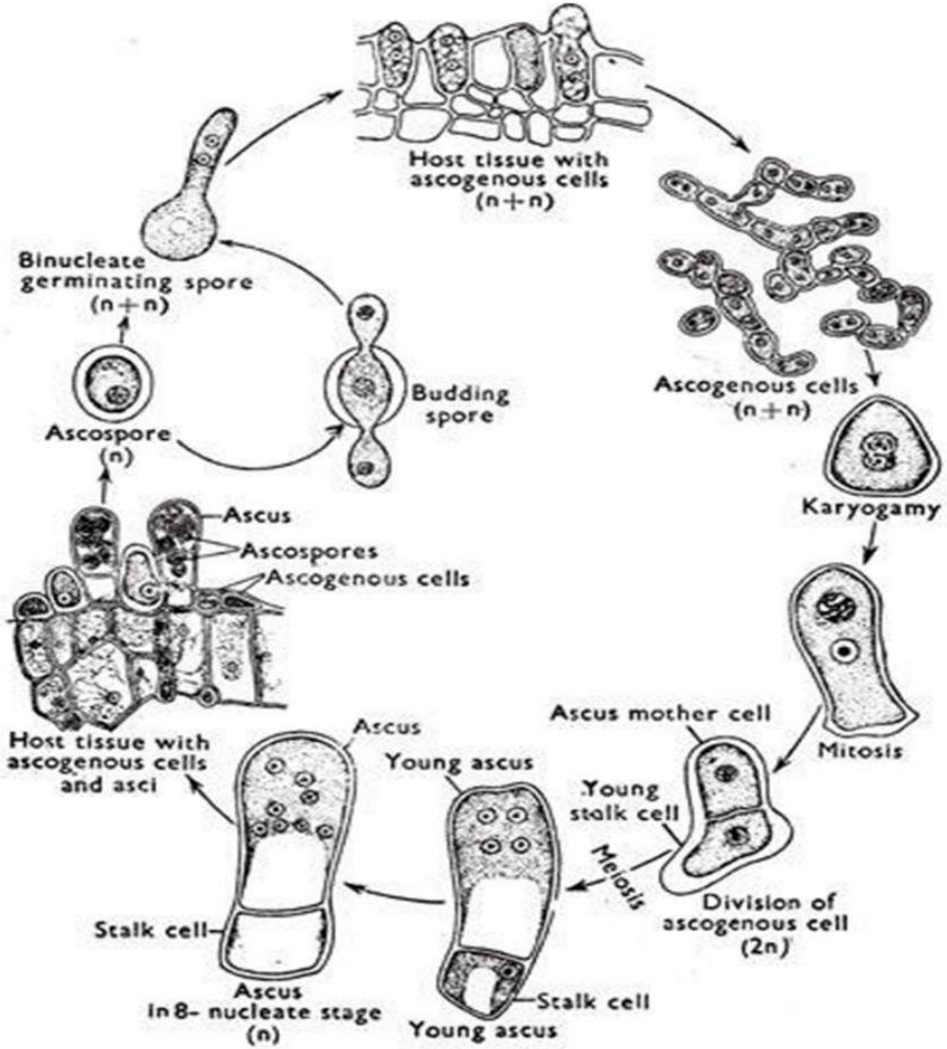
- The spots coalesce freely and form irregular lesions and leaves ultimately dry.
- The infected leaves are **distorted**, have a **reddish brown appearance** and become yellow much sooner than the healthy leaves.
- The disease hastens defoliation of leaves.

Taphrina leaf blotch - *Taphrina maculans*



- The fungus produces cuboid ascogenous cells several layers in depth in the subcuticular interspaces of the epidermis.

Taphrina leaf blotch - *Taphrina maculans*

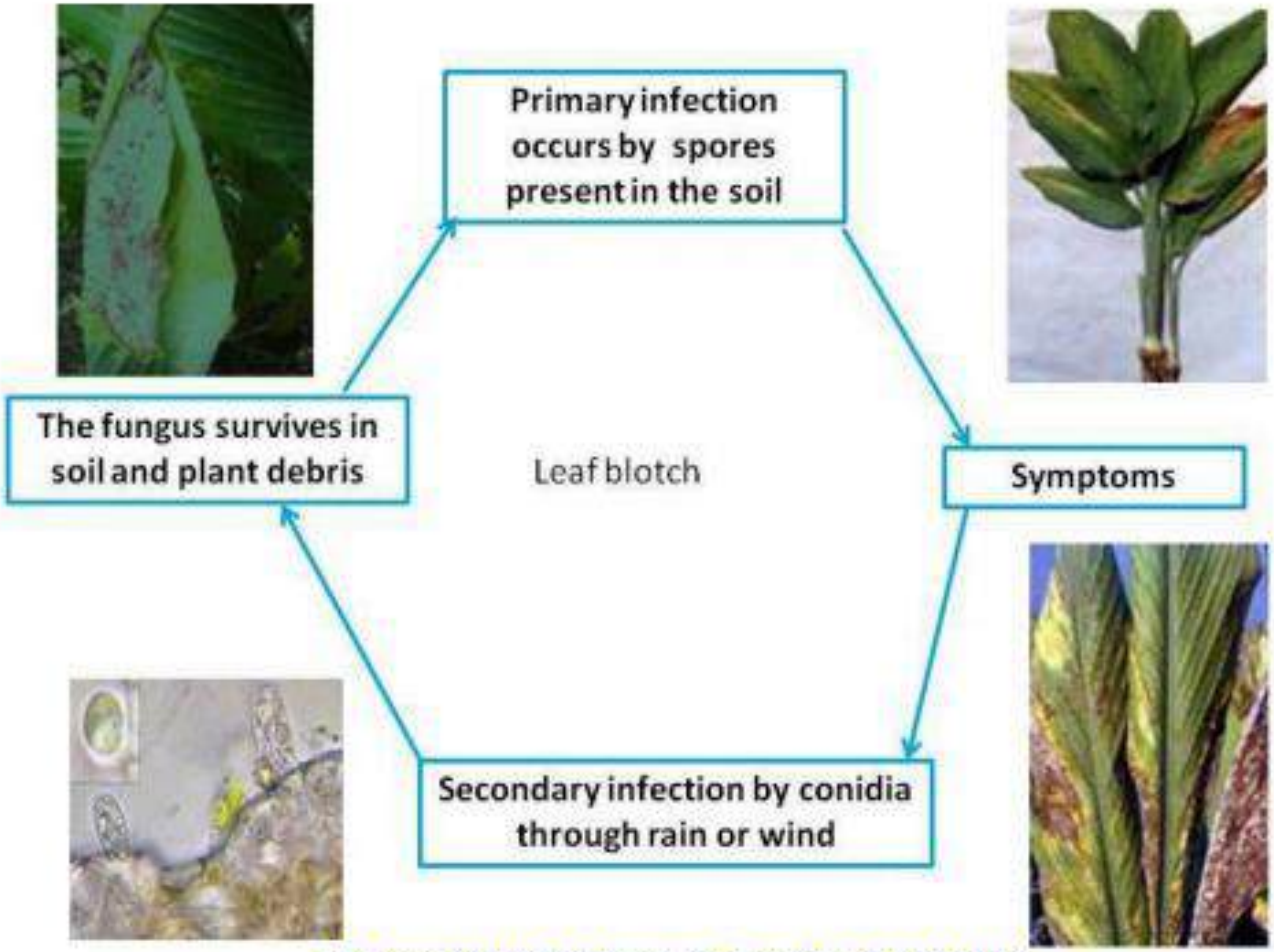


Taphrina leaf blotch - *Taphrina maculans*

Disease cycle and epidemiology:

- The fungus persists by means of **ascogenous cells on the leaf debris** and as **dessicated ascospores and blastospores** in the soil and among the fallen leaves.
- Ascospores cause primary infections.
- Optimum temperature for infection is 21-23° C along with 80 per cent RH.

Taphrina leaf blotch - *Taphrina maculans*



http://www.ikisan.in/Crop%20Specific/Eng/Images/fur_dm_leafblotch.jpg

Taphrina leaf blotch - *Taphrina maculans*

Management:

- Collection and destruction of the infected plant debris.
- Resistant cvs. **like CLL 324, Amalapuram, Mydukur, Karhadi local, CLL326, Ochira 24 & Alleppey**.
- With the initiation of the disease, spraying with fungicides like carbendazim (0.1%) or thiophanate methyl (0.1%) or mancozeb (0.25%) or propineb (0.25%) and repeat at 10-14 days interval.

Leaf spot or anthracnose - *Colletotrichum capsici*

elliptic or oblong spots

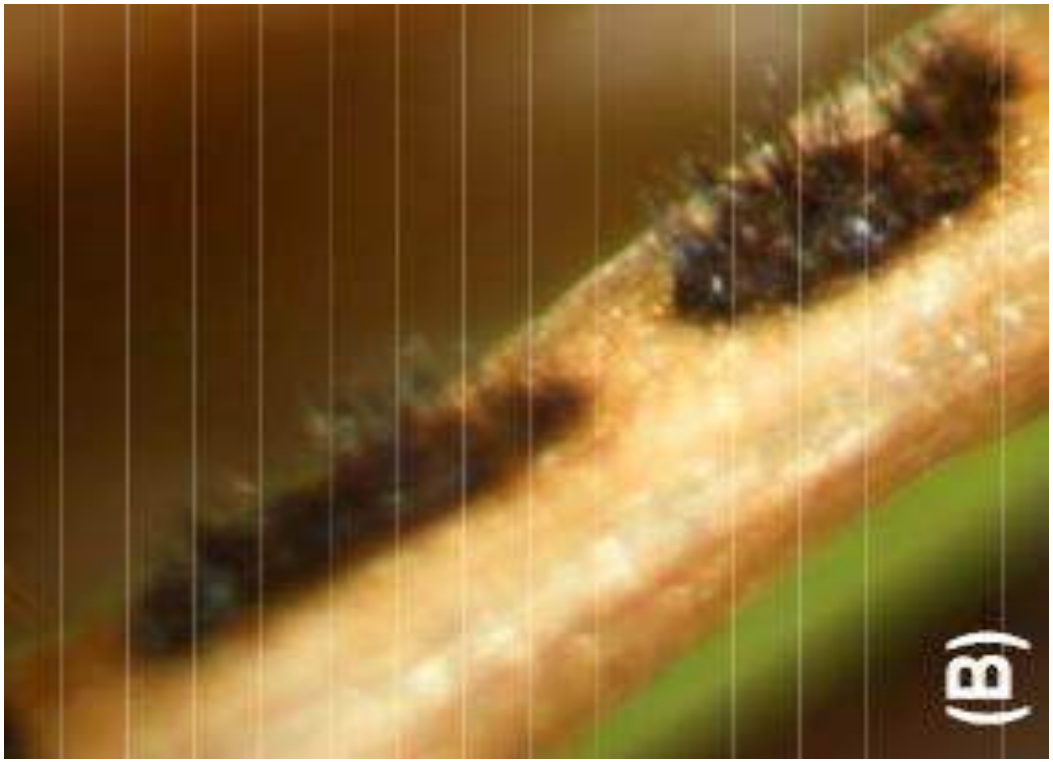
Yellow halo

Centre of the spot is greyish white and thin

spots coalesce and develop irregular patches

Leaf spot or anthracnose - *Colletotrichum capsici*

The infected field presents a scorched look



Centre of the spot is greyish white and thin with numerous black dot like acervuli

Leaf spot or anthracnose - *Colletotrichum capsici*

- This disease was first reported from Coimbatore districts in Tamil Nadu and is now prevalent in all the turmeric growing tracts in India.

- The disease mostly appears during **August and September when the humidity is high.**

Symptoms:

- The disease manifests in the form of elliptic or oblong spots of variable size.

- In the initial stages, the spots remain small but very soon many of them **increase in size**.

Leaf spot or anthracnose - *Colletotrichum capsici*

Symptoms contd…

- Many **spots coalesce** and develop irregular patches often involving a major portion of the leaf which **eventually dries up.**
- The centre of the spots is **greyish white and thin with numerous black dot like acervuli** on both surfaces.
- There is an indefinite yellowish region outside the spot forming a **halo** around.
- The infected field presents a **scorched look** during the years of epiphytotics.

Leaf spot or anthracnose - *Colletotrichum capsici*

Pathogen:

Colletotrichum capsici also pathogen of bell pepper and chilli

Disease cycle and epidemiology

- **Survives in infected plant residues and diseased rhizomes**
- The pathogen may be carried over in the rhizomes as **dormant stromata between the scales.**
- The disease appears when there is high and continuous humidity in the atmosphere.

Leaf spot or anthracnose - *Colletotrichum capsici*

Management:

- Healthy seed rhizomes.
- Resistant cvs. **like Nallkatla, Sugandham, Duvvur and Ganikota.**
- With the initiation of the disease, spray the crop with **Mancozeb (0.25%) or mancozeb (0.25%) + carbendazim (0.1%) or tricyclazole (0.03%) and repeat at 10 -14 days interval.**

Rhizome rot or root rot - *Pythium aphanidermatum, P. graminicolum & Fusarium solani*

- Progressive drying of the leaves, which proceeds first along the margins and later the entire leaf dries up.

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Rhizome rot or root rot - ***Pythium aphanidermatum, P. graminicolum & Fusarium solani***

- The base of the aerial shoots shows watersoaked soft lesions.

Rhizome rot or root rot - *Pythium aphanidermatum, P. graminicolum & Fusarium solani*

- **Infection gradually spreads to rhizomes which begin to rot and become soft**
- **Rotted rhizome can be easily pulled out**
- **The bright orange colour of the rhizome is changed into different shades of brown**
- **The rotten rhizomes emit foul smell during advanced stages**

Rhizome rot or root rot - *Pythium aphanidermatum, P. graminicolum & Fusarium solani*

- The yield is considerably reduced due to the attack of this disease.

Symptoms:

- The affected plants show progressive drying of the leaves, which proceeds **first along the margins and later the entire leaf dries up.**
- The **base of the aerial shoots shows watersoaked soft lesions.**
- The root system is adversely affected and as the disease progresses, **infection gradually passes to the rhizomes**, which begin to rot and become soft.

Rhizome rot or root rot - *Pythium aphanidermatum, P. graminicolum & Fusarium solani*

Symptoms contd…

- The colour of the affected rhizomes is changed into **different shades of brown**.

Pathogen

Similar to ginger

Disease cycle and epidemiology

Similar to ginger

- These fungi are both **seed and soil borne**.
- High temperature and soil moisture favours the disease development.

Rhizome rot or root rot - *Pythium aphanidermatum, P. graminicolum & Fusarium solani*

- Select healthy seed rhizomes for sowing.
- Collect and destroy the infected clumps and burn them.
- Follow atleast three years crop rotation.
- Tolerant cvs. like PCT-13, PCT-14, and Shillong.
- Dip seed rhizomes in the solution of metalaxyl + mancozeb (0.25%) before sowing and with the initiation of the disease, drench the plants with the same chemical.

Diseases of ginger

1. Phyllosticta leaf spot - *Phyllosticta zingiberi*

2. Rhizome rot or soft rot - *Pythium aphanidermatum*, *P. butleri*, *P. complectens*, *P. deliense*, *P. gracile*, *P. graminicolum*, *P. myriotylum*, *P. pleroticum*, *Fusarium solani* and *F. equiseti*.

3. Banded Leaf and Sheath Blight - *Rhizoctonia solani*

4. Bacterial wilt - *Ralstonia solanacearum*

Phyllosticta leaf spot - *Phyllosticta zingiberi*

- **On leaves, small oval to elongated spots appear which later on develop white papery centre and dark brown margins with a yellow halo**

Phyllosticta leaf spot - *Phyllosticta zingiberi*

Symptoms:

- On leaves, **small oval to elongated spots** appear which later on develop **white papery centre** and **dark brown margins with a yellow halo.**

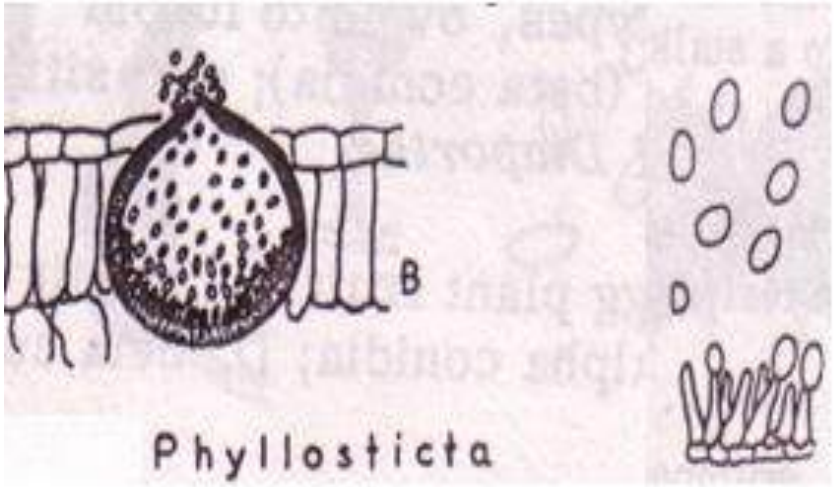
- The affected leaves suffer from extensive desiccation and are shredded and disfigured.

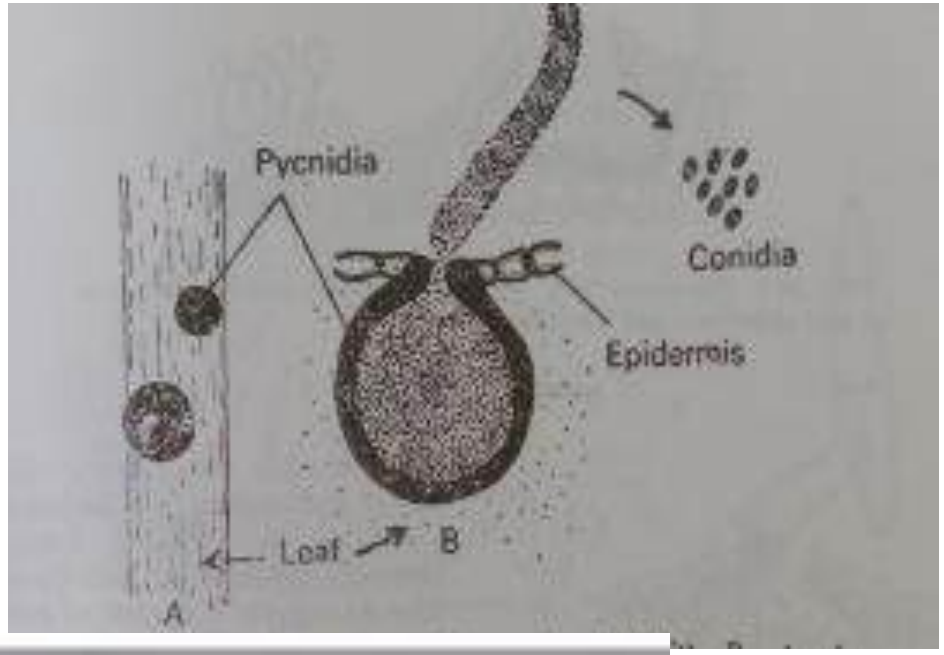
Pathogen: *Phyllosticta* (teleomorph: *Guignardia***)**

- The disease is caused by *Phyllosticta zingiberi* .
- The fungus forms amphigenous, subglobose, dark brown, ostiolate pycnidia on the host.
- Pycnidiospores are hyaline, unicellular, oblong and biguttulate.

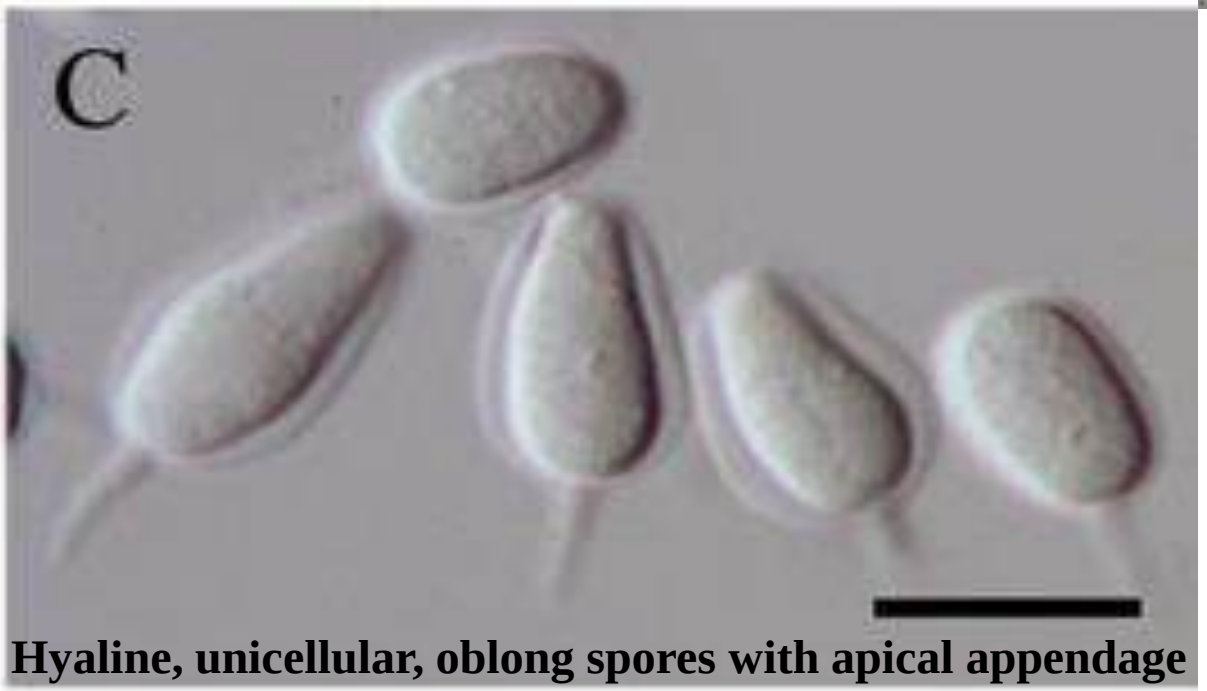
Phyllosticta leaf spot - *Phyllosticta zingiberi*

White papery centre and dark brown margins with pycnidia in the centre



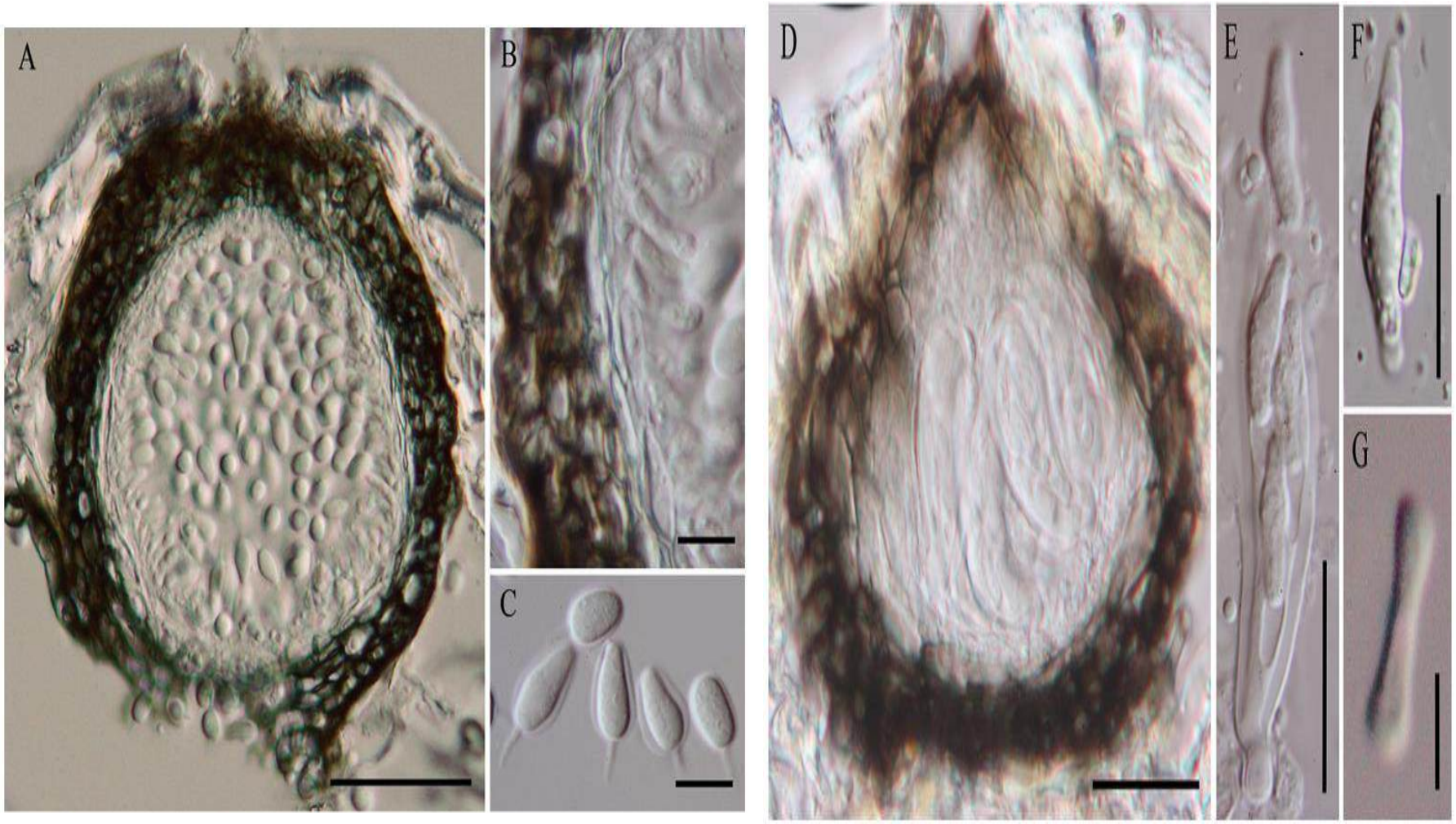


Pycnidia - dark, ostiolate, globose



Hyaline, unicellular, oblong spores with apical appendage

Pycnidia **Ascostroma**

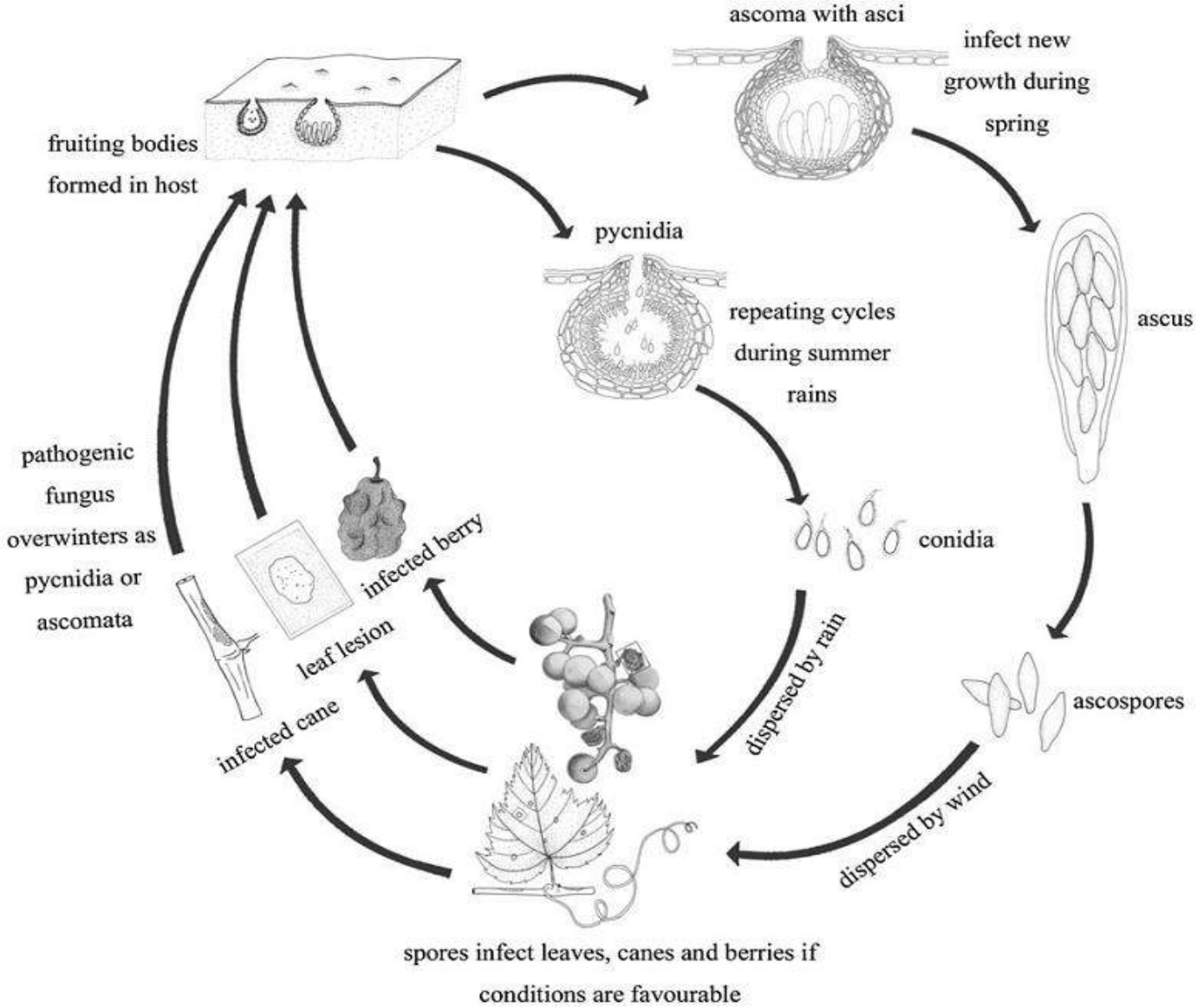


Guignardia (family *Phyllostictaceae* in *Botryosphaeriales*) produce simple brown, thin-walled ascomata, containing **bitunicate and fissitunicate asci** and variously shaped, hyaline, unicellular ascospores, with or without mucilaginous sheaths or appendages

Phyllosticta leaf spot - *Phyllosticta zingiberi*

Disease cycle and epidemiology:

- The fungus overwinters in **infected plant debris and seed rhizomes.**
- The disease **begins to appear towards the end of June** when temperature varies from 23.4 to 29.6 and RH is in between 83.3 to 90.2 per cent.
- Later, **in July** when rainy days and total rainfall increases, the disease **aggravates and spread fast**.
- Higher intensity of rain, accompanied by wind, has greater impact on target leaves and **fungus is splashed to a greater distance resulting in increased disease incidence**



g. 2 Schematic representation of the life cycle of *Phyllosticta* and its teleomorph redrawn from: http://www.oardc.ohio-state.edu/fruitpathol ganic/grape/All-Grapes.html

Phyllosticta leaf spot - *Phyllosticta zingiberi*

Management:

- Collection and destruction the plant debris by burning.
- By dipping healthy seed rhizomes in the mixture of mancozeb (0.25%) and carbendazim (0.1%) for 60 minutes before sowing and dry in shade for 48 h.
- Spray the crop with mancozeb (0.25%) or carbendazim (0.1%) or combination of mancozeb (0.25%) and carbendazim (0.05%) and repeat at 10 to 14 days interval.
- Moderately resistant cultivars like **Tura, Nadia, Tetraploid and Thingpani**.

Rhizome rot or soft rot - *Pythium aphanidermatum, P. butleri, P. complectens, P. deliense, P. gracile, P. graminicolum, P. myriotylum, P. pleroticum, Fusarium solani* and *F. equiseti.*

- The initial symptoms of the disease are noticed on leaves which turn slightly pale.
- The yellowing of the leaves starts from the tip of the blade and spread downwards

Rhizome rot or soft rot - *Pythium aphanidermatum, P. butleri, P. complectens, P. deliense, P. gracile, P. graminicolum, P. myriotylum, P. pleroticum, Fusarium solani* and *F. equiseti.*

Symptoms:

- Two types of symptoms i.e. wet and dry rot are generally noticed on rhizomes.
- The initial symptoms of the disease are noticed on leaves which turn slightly pale.
- The yellowing of the leaves starts from the **tip of the blade and spread downwards.**
- **Ultimately the infected leaves are killed which droop and hang down along the pseudostem**
- The basal portion of the plant exhibits pale translucent colouration which becomes water soaked and soft

Rhizome rot or soft rot - *Pythium aphanidermatum, P. butleri, P. complectens, P. deliense, P. gracile, P. graminicolum, P. myriotylum, P. pleroticum, Fusarium solani* and *F. equiseti.*

Symptoms contd…

- When infection takes place through contaminated seed rhizomes, sprouts fail to grow resulting in pre-emergence damping off
- The infection starts at the collar region of the pseudostems and progresses upwards as well as downwards.
- Plants become pale, leaf tips turn yellow and the infection gradually spreads down the leaf blade and leaf sheath along the margin.
- Infected plants can be pulled out easily collar region becomes soft, rotten and show pale, translucent, water soaked patches;

Rhizome rot or soft rot - *Pythium aphanidermatum, P. butleri, P. complectens, P. deliense, P. gracile, P. graminicolum, P. myriotylum, P. pleroticum, Fusarium solani* and *F. equiseti.*

Symptoms contd…

- Rotting spreads to the rhizome resulting in soft rot.
- Lesions gradually decompose forming a watery mass of putrifying tissues enclosed by the tough rind of the rhizome and the fibrovascular strands are not affected.
- Roots arising from the affected regions of rhizome also show typical softening and rotting
- Rotting is also noticed in rhizomes collected from infected crop during storage

Rhizome rot or soft rot - *Pythium aphanidermatum, P. butleri, P. complectens, P. deliense, P. gracile, P. graminicolum, P. myriotylum, P. pleroticum, Fusarium solani* and *F. equiseti.*

Diseased ginger rhizomes

- The rotting extends from collar regions to rhizomes.

Rhizome rot or soft rot - *Pythium aphanidermatum, P. butleri, P. complectens, P. deliense, P. gracile, P. graminicolum, P. myriotylum, P. pleroticum, Fusarium solani* and *F. equiseti.*

Pathogen (s):

- The disease is caused by different species of *Pythium* and *Fusarium* amongst these importantare *P. aphanidermatum*, *P. butleri* , *P. complectens*, *P. deliense*, *P. gracile* , *P. graminicolum* , *P. myriotylum*, *P. pleroticum*, *Fusarium solani* and *F. equiseti*.
- The hyphae of *P. aphanidermatum* are hyaline, branched, non septate which form long tapering sporangia by swelling of the hyphae.
- The sporangium blows out into a vesicle to which the protoplasm migrates.
- **Zoospores** are formed in vesicle and are kidney shaped, biflagellate and slightly depressed at the hilum end.
- The **oospores** are smooth walled, plerotic and spherical in shape.
- ***Fusarium equiseti* produces micro and macro conidia.**
- Micro conidia are aseptate whereas macro conidia are 1-3 septate, sickle to spindle shaped.

Rhizome rot or soft rot - *Pythium aphanidermatum, P. butleri, P. complectens, P. deliense, P. gracile, P. graminicolum, P. myriotylum, P. pleroticum, Fusarium solani* and *F. equiseti.*

Disease cycle and epidemiology:

- The pathogen (s) perpetuate through diseased rhizomes as well as through oospores.

- The infected plant debris remaining in the field serves as primary source of infection.

- Oospores have also been detected in the scales of stored rhizomes.

Rhizome rot or soft rot - *Pythium aphanidermatum, P. butleri, P. complectens, P. deliense, P. gracile, P. graminicolum, P. myriotylum, P. pleroticum, Fusarium solani* and *F. equiseti.*

Disease cycle and epidemiology:

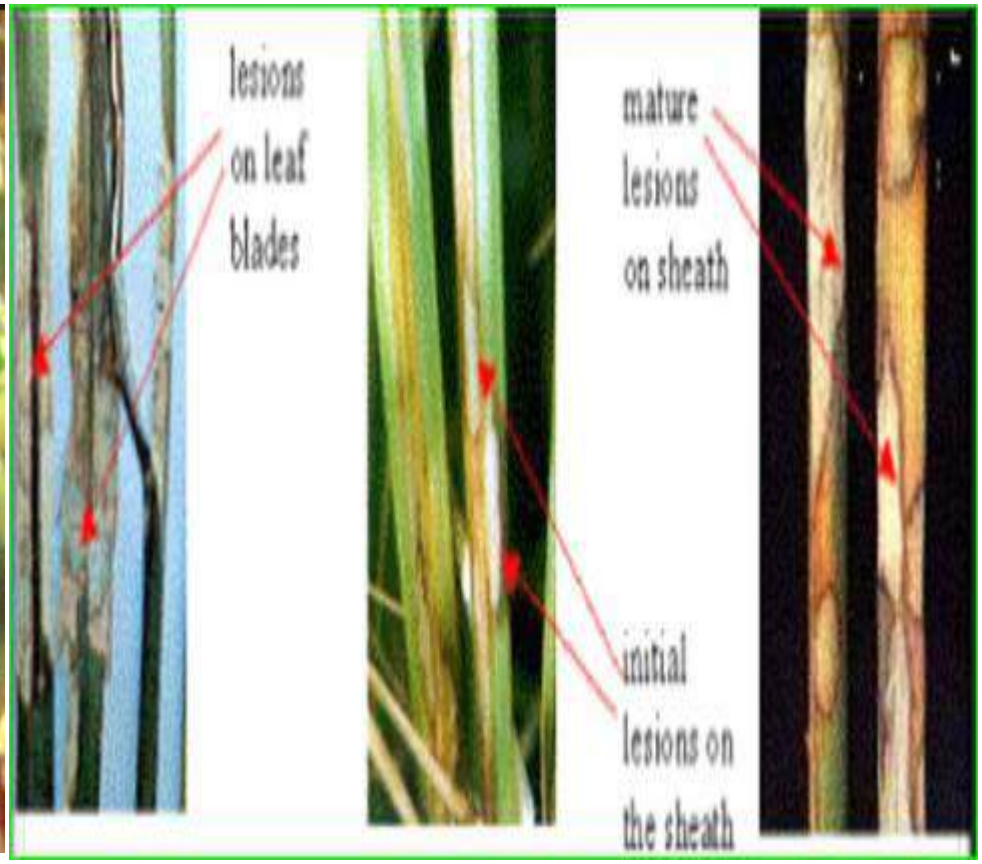
- A warm and humid climate predisposes the plants to infection at sprouting stage.
- The spread of the disease is typical of soil borne diseases.
- For germination of *Pythium* spp. like *P. aphanidermatum* and *P. myriotylum*, the optimum temperature is 34° C and for *P. pleroticum* is $25\text{-}30^{0}$ C whereas for growth and multiplication of *F. equiseti*, the optimum temperature is 30° C.

Rhizome rot or soft rot - *Pythium aphanidermatum, P. butleri, P. complectens, P. deliense, P. gracile, P. graminicolum, P. myriotylum, P. pleroticum, Fusarium solani* and *F. equiseti.*

Management:

- Use disease free seed rhizome for planting and treat by steeping them in the mixture of mancozeb (0.25%) and carbendazim (0.1%) for 60 minutes before storage as well as sowing followed by drying in shade for 48 hours.
- Follow at least 2-3 years crop rotation.
- Apply neem cake (250 g/m^2) and lime as soil amendment.
- The antagonistic species of *Trichoderma* such as *T. viride*, *T. harzianum* and *T. hamatum* are known to inhibit the growth of the pathogenic fungi (40 g/m2).
- Drench the affected plants with copper oxychloride (0.3%) or Bordeaux mixture (4:4:50) during rainy season.

Banded Leaf and Sheath Blight - *Rhizoctonia solani*



Lesions are small, ellipsoid or ovoid, and greenish-gray - near the water line in lowland fields

Banded Leaf and Sheath Blight - *Rhizoctonia solani*

Symptoms:

- The lesions are usually observed on the leaf sheaths and leaf blades
- Initially, lesions are small, ellipsoid or ovoid, and greenish-gray and usually develop near the water line in lowland fields.
- Under favorable conditions, they enlarge and may coalesce forming bigger lesions with irregular outline and grayish-white center with dark brown borders.
- The presence of several large spots on a leaf sheath usually causes the death of the whole leaf
- Large discoloured areas develop alternating with irregular bands.
- Blotching of leaf and leaf sheath occurs in case of severe infection

Banded Leaf and Sheath Blight - *Rhizoctonia solani*

Disease cycle

- *Rhizoctonia solani* survives in the soil and on infected crop debris as sclerotia or mycelium.
- Sclerotia are known to survive for several years in the soil.
- The fungi spread by water (flooding), irrigation, movement of contaminated soil, and plant debris.

Banded Leaf and Sheath Blight - *Rhizoctonia solani*

Management – Prophylactic

•Seed treatment with *Pseudomonas fluorescens* @ of 2.5 kg or products/ha dissolved in 100 litres and dipping for 30 minutes.

•Soil application of *P.fluorescens* @ of 2.5 kg/ha after 30 days of transplanting (This product should be mixed with 50 kg of FYM/sand and then applied.

•Foliar spray of *P. fluorescens* at 0.2% concentration ,commencing from 45 days after transplanting at 10 days interval for 3 times depending upon the intensity of disease.

Banded Leaf and Sheath Blight - *Rhizoctonia solani*

Management – Cultural

- Apply FYM 12.5 t/ha or green manure 6.25 t/ha to promote antagonistic microflora
- Avoid excess doses of fertilizers.
- Adopt optimum spacing.
- Eliminate weed hosts.
- Apply organic amendments.
- Avoid flow of irrigation water from infected fields to healthy fields.
- Deep ploughing in summer and burning of stubbles.

Management – Curative

- Spraying with Carbendazim (1 g/lit), Propiconazole (1ml/lit) Chlorothalonil 4 g/lit

Bacterial wilt - *Ralstonia solanacearum*

Symptoms:

- Water-soaked patches or streaks appear on collar region which slowly enlarge.
- Bronze colouration on leaf margins is also observed and the leaves slowly become flaccid with intense yellowish bronze colour and ultimately droop down exhibiting typical wilt symptoms.
- At advanced stages, pseudo stem appears slimy and if infected rhizomes are pressed, a milky bacterial exudates oozes out in clear water

Bacterial wilt - *Ralstonia solanacearum*

Water-soaked patches or streaks appear on collar region which slowly enlarge

Leaves slowly become flaccid with intense yellowish bronze colour and ultimately droop down

Bacterial wilt - *Ralstonia solanacearum*

Bacterial wilt - *Ralstonia solanacearum*

Pathogen:

- *Ralstonia solanacearum* Similar to tomato

Disease cycle and epidemiology:

- The bacterium perpetuates through infected rhizomes and soil. Biotype III of the bacterium is known to cause this disease in India.
- The bacterium has wider host range including both cultivated and weeds.
- The presence of nematodes has a positive and significant effect on the development of bacterial wilt.

Bacterial wilt - *Ralstonia solanacearum*

Management:

- Use healthy seed rhizomes for planting.
- Eradicate weed hosts and adopt effective crop rotation for at least three years.
- Treat seed rhizomes with Streptocycline (100 ppm) for 30 minutes before planting and spray the crop with Streptocycline (100 ppm) starting from 1 month after transplanting and repeat at fortnightly interval.